

MBSD Assignment #3 A.Y. 2024/25

Purposes

- Perform some parts of the Functional and Technical Safety Concept analysis, according to ISO26262, of a “one pedal controller” for a car.
- Implement some of the safety concepts in the Simulink model of the controller developed in Laboratory #2.
- Perform unit and integration tests on the implemented safety-related functionalities.

It is available an example of a Functional Safety Concept for the item Front Light Manager (FLM).

The deliverable, composed of

- the report (the following pages of this document)
- the Simulink models on where the safety concepts have been implemented
- all the needed files to replicate the software testing results

has to be provided as a .ZIP file (do not use other compression formats) up to **May 14th at 23:59**. It shall also contain a brief report explaining the design of the controller using the following template.

It is sufficient that only one of the group members uploads it.

IMPORTANT:

For making the Functional Safety Concept coherent between the groups of the course, consider as ASIL C all the safety goals related to unintended acceleration (those leading to an increase of the vehicle's speed modulus) and as ASIL B the warnings to the driver and the unintended deceleration (those leading to a decrease of the vehicle's speed modulus).

Model-Based Software Design, A.Y. 2022/23

Laboratory 3 Report

Components of the working group (max 2 people)

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Functional Safety Concept

One pedal

Functional safety architecture

Figure 1 Functional safety architecture

Attributes of the Safety Goals

Fill the table with the safety goal descriptions

SG 1	Correctly switching to a safe state once a pedal position error is detected.
SG 2	Notifying the user when CAN errors occur and limiting the torque requested to a null value.
SG 3	

Fill in the attribute/parameters of the safety goal

Safety goal	Attributes/Parameters of the safety goal				
	Integrity (ASIL)	Safe state	Fault tolerance time	Warning concept	Degradation concept
SG1	ASIL C	Neutral	100ms	Notify driver in case of error and automatically handle the state switching	Red display on screen
SG2	ASIL C	Neutral	50 ms	State change and user- error communication	

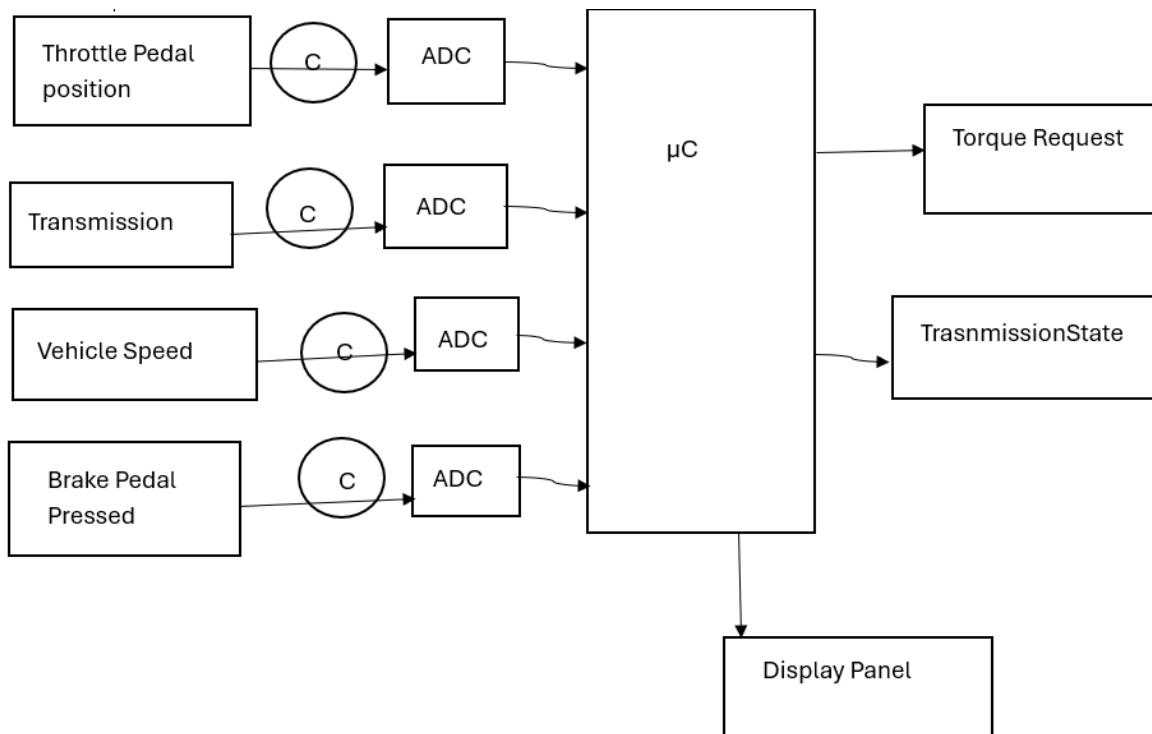
Functional (and technical) safety requirements and allocation

	Define functional safety requirements		Allocation of requirements on systems and elements	
	Safety requirements	Remark	If applicable, allocate the safety requirements to other Items / Systems	If applicable, allocate the safety requirements to equipment other technologies to minimize risk. That could be e.g. hydraulic, mechanical equipment

Safety goals	SG1	The throttle pedal position is measured an additional time by a redundant sensor to compare to the default one	N/A	Alarm sound	
	SG2	Once the system runs the communication from the CAN signal is identified based on the input		Alarm sound	

ASIL preliminary architecture¹

The purpose of this figure is to assign ASILs to the components indicated in Figure 1 on the functional safety architecture.



¹ See document 02-iso26262.pdf, slides 89, 90, 91, 92, 93.

Implementations²

Functional redundancies

Indicate which components of the functional safety architecture are implemented with redundancies.

The additional pedal position sensor is inevitable if such a failure is to be detected. For that purpose, this redundant component has been adopted to compare with in order to determine the true value under the hypothesis of single error.

Implemented plausibility checks

Indicate which signals are subjected to plausibility checks. Indicate also the checks needed for the channels with redundancies in place.

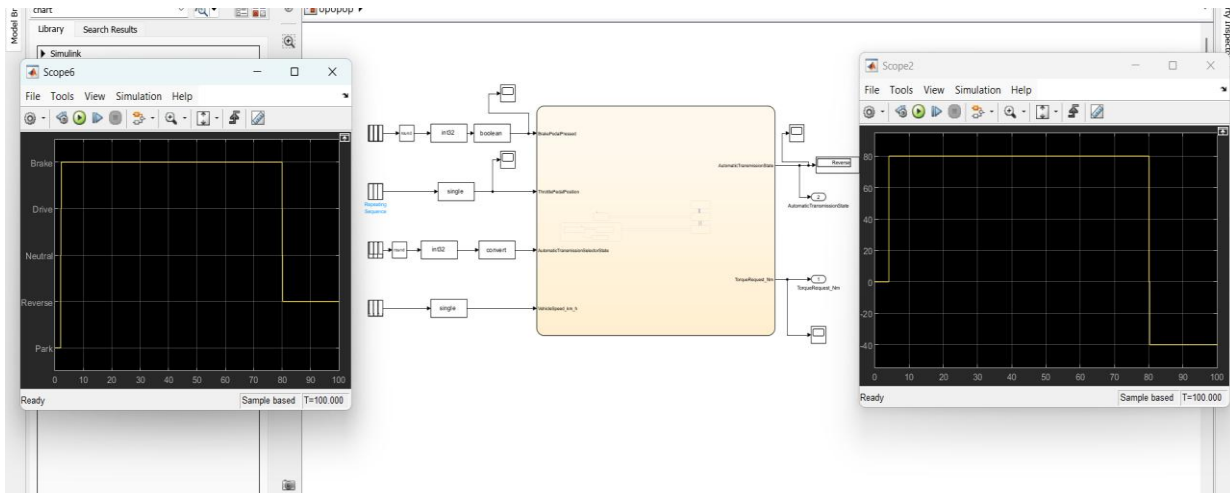
The can signal and the pedal position are subjected to plausibility checks. For that purpose, a state flow chart has been to signal the controller to take into consideration such a check before execution of task.

² In the ISO26262 the implementations are based on a document called *Technical Safety Concept*, but for simplicity we move straight from the *Functional Safety Concept* to software implementations. A guideline for the implementation phase can be found in the document *02-iso26262.pdf* from slide 81, in particular slide 86.

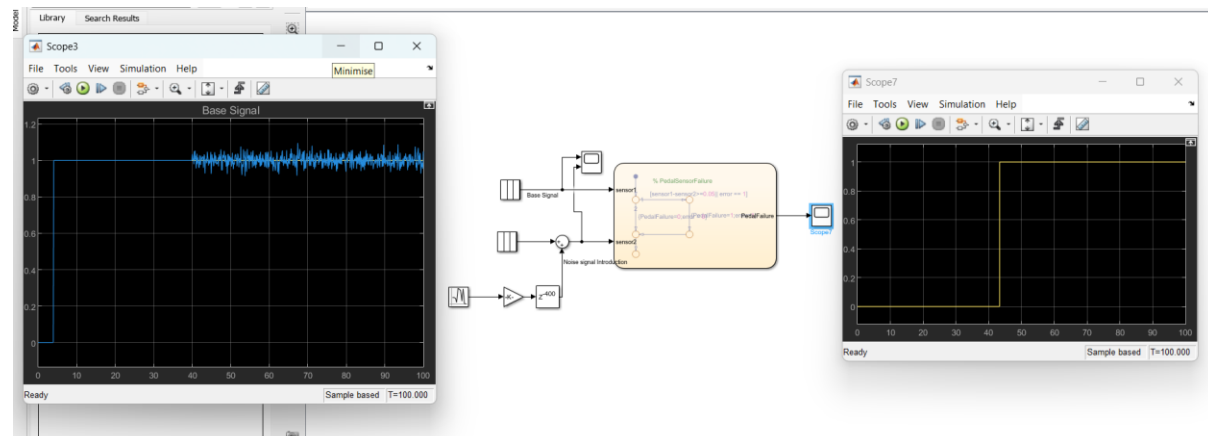
Software testing

Implemented unit tests

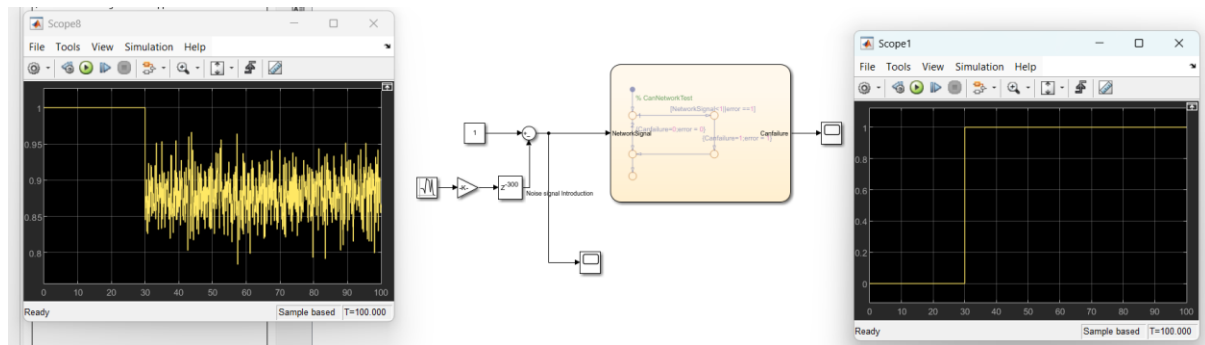
Describe in English the test performed to verify the correct functionality of the safety mechanism implemented.



In this figure the unit test of the controller has been tested to verify its functionality. A certain input has been fed for the controller and the expected output has been verified as indicated above. To the left is the transmission state at the output which is coherent with the input we provided and alternatively the requested torque to the right has been verified with the expected values for reverse, drive and brake.



For the case of the pedal position to simulate what would happen in a real case a noise signal has been to disturb the signal, following the block chart an error should occur at the instance where the error from the sensor 1 is different from that of sensor 2, thus validating our purpose.

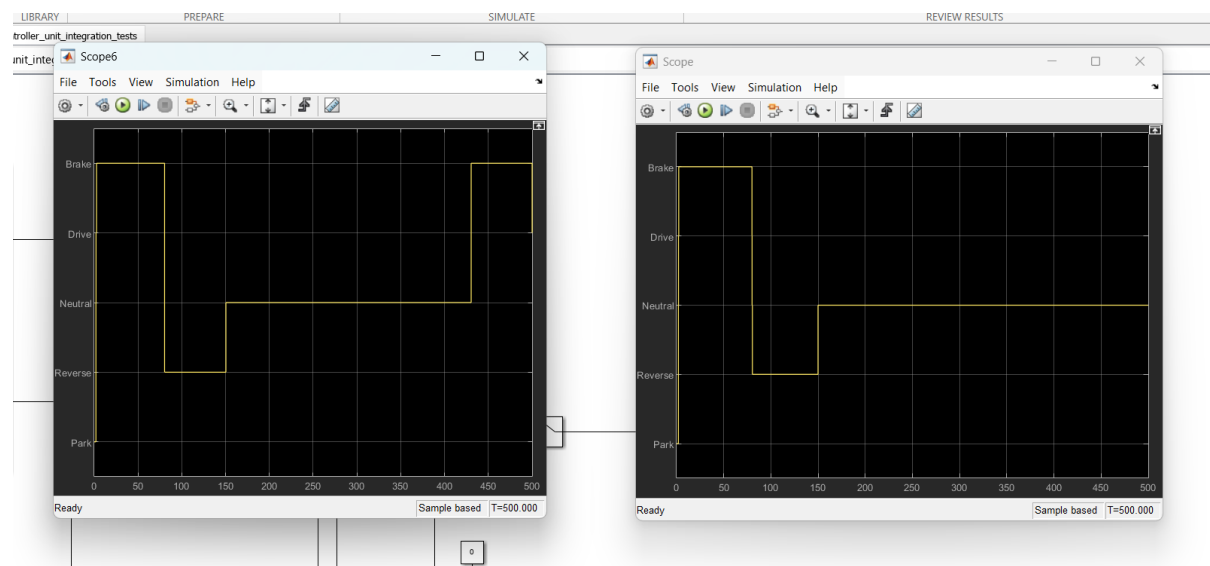


Similarly in the case for the CAN network a disturbance has been introduced where the digital signal of the network is disturbed and thus triggering a safe state.

Implemented integration tests

Describe, in English, the scenarios tested at the integration level to verify the proper integration between the various units implementing the safety mechanisms.

For the integration test, all the controller has been mounted together and the safety mechanism has been tested.



To test the hypothesis, a disturbance has been introduced at time = 150 ms of the simulation time. The figure on the left shows the output of the controller without the safety mechanism however the one on the right shows the transmission ratio that is fed to the plant to be neutral in addition to a zero torque request.